

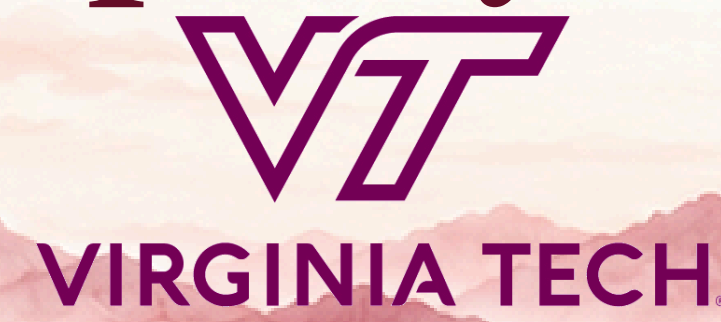
Capacity-Dependent Effects of Data Selection for Mathematical Reasoning

High-Likelihood Data Is Not Always Better

Cuong Dang, Hoang Anh Just, Ruoxi Jia

Department of Electrical and Computer Engineering Virginia Tech, Blacksburg, VA, USA
{cuongdc, just, ruoxijia}@vt.edu

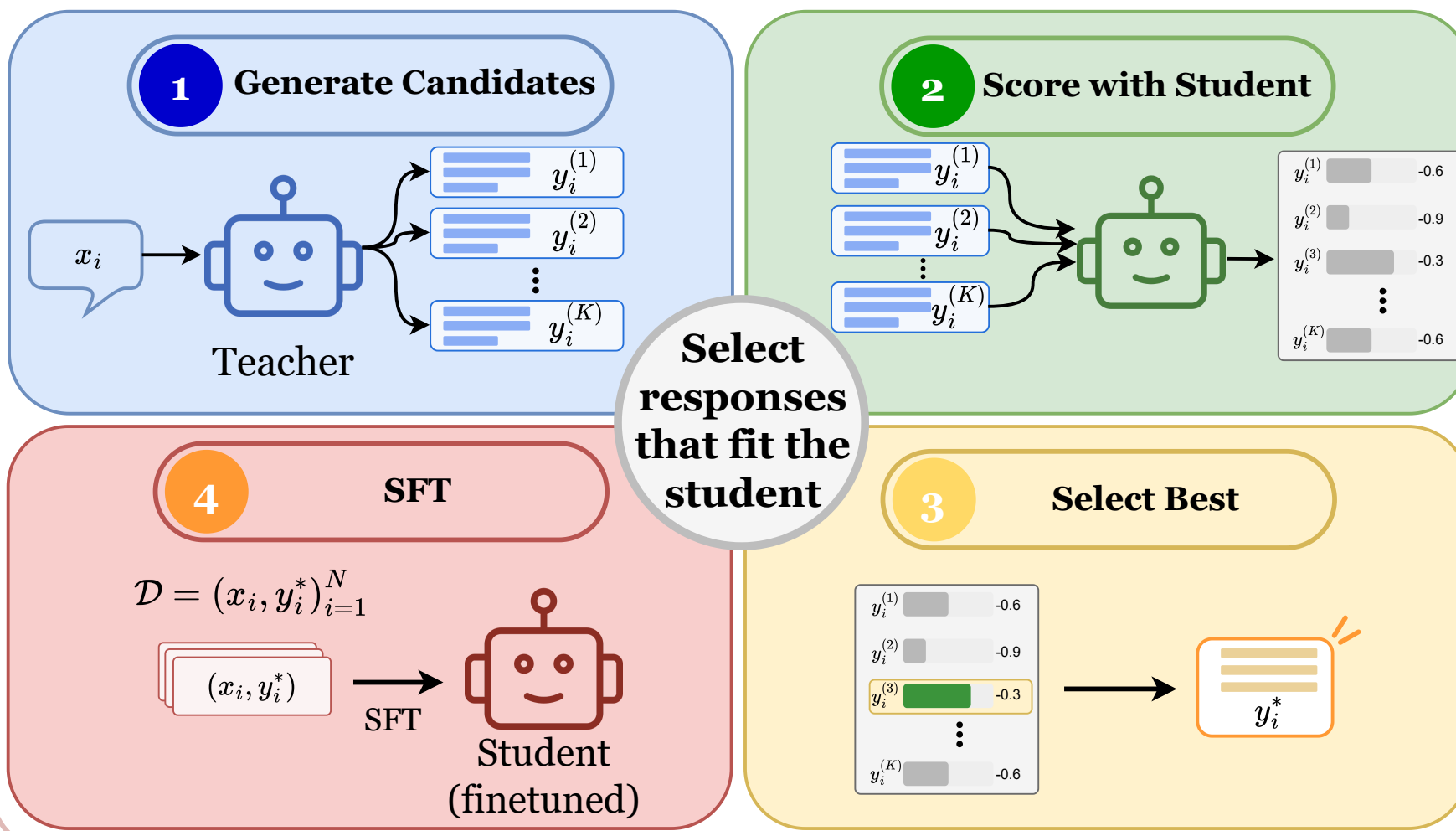
COLM 2026
San Francisco, CA, US



1. Motivation / Questions

Prior likelihood-based response selection often favors high-likelihood responses.

Is high-likelihood data always best?



2. Experimental Setup

Students	1.5B to 8B models across Qwen and Llama families.
Data	Teacher-generated reasoning responses on MATH12K.
Selection	High-likelihood vs low-likelihood examples based on student likelihood.
Training	Compare early training and longer training.
Evaluation	Pass@8 on 10 reasoning benchmarks: AIME, AMC, CHMATH, GAOKAO, GPQA, Grade School, KAOYAN, Minerva, Olym. Bench

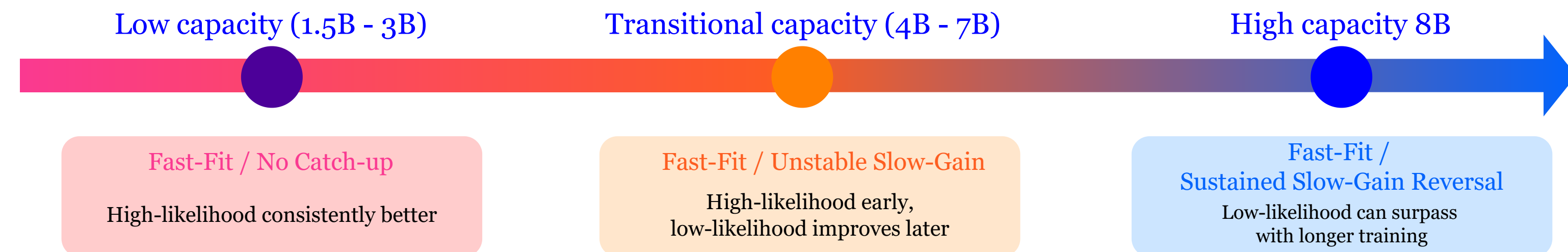
3. Core Idea: Different Models, Different Learning Dynamics



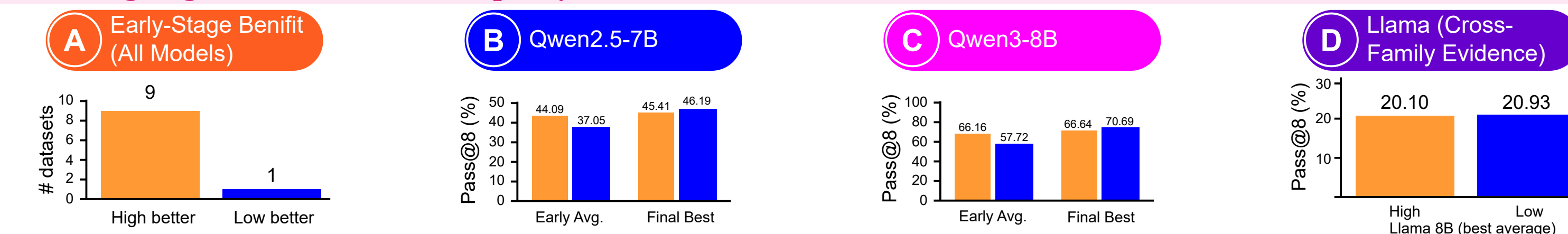
4. Key Empirical Findings

- Small models under low-likelihood training often repeat or get lost; larger models can eventually reason correctly.
- High-likelihood data gives faster early gains, especially for smaller models.
- For larger models, low-likelihood data improves later and can surpass high-likelihood.

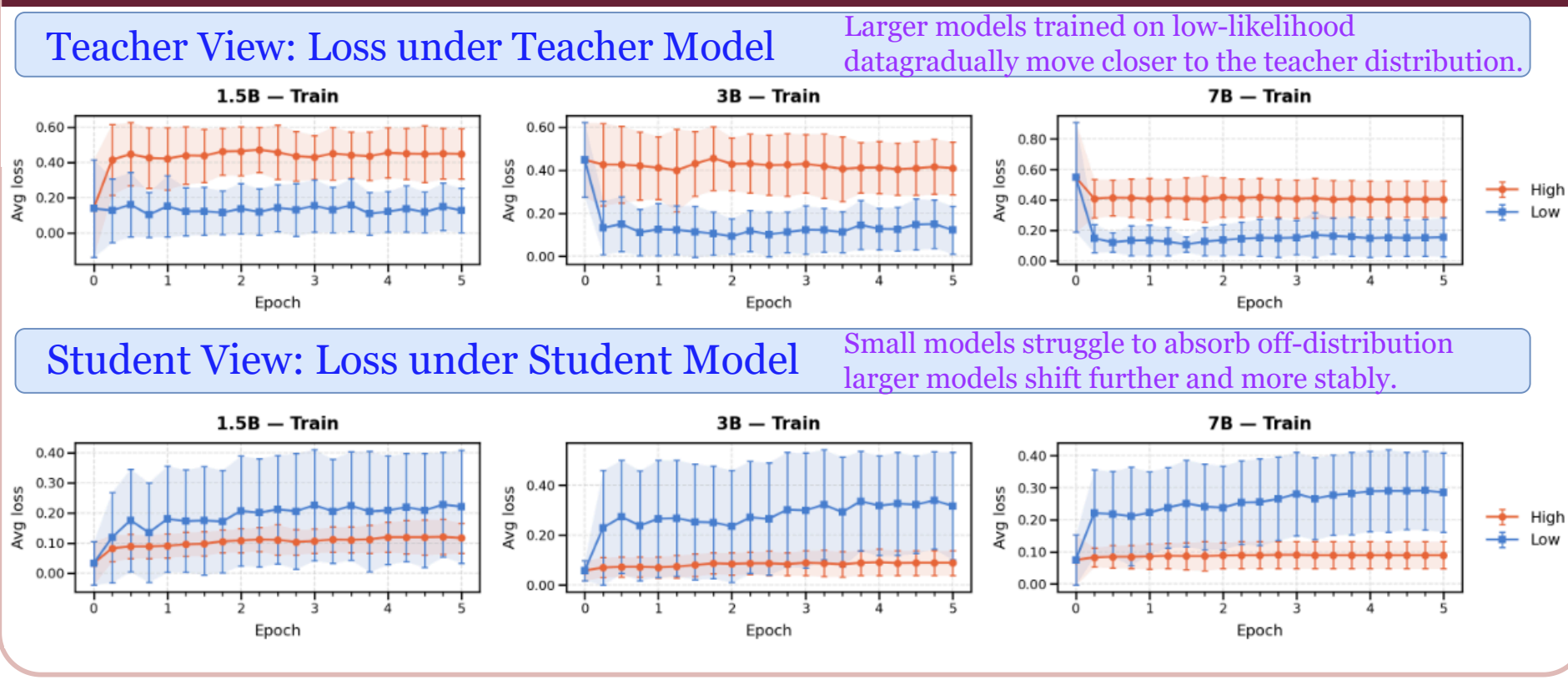
Learning Regimes Across Model Capacity



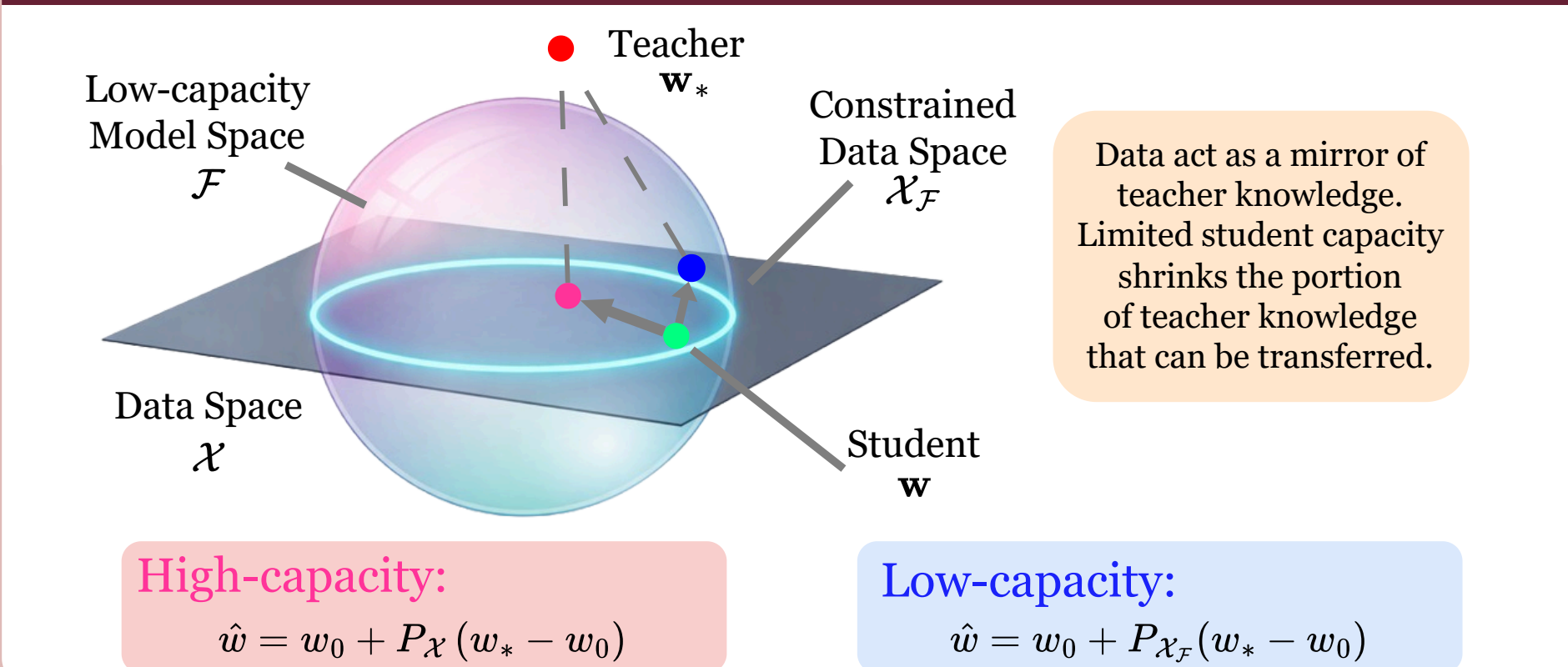
Learning Regimes Across Model Capacity



5. Dual-view Learning Dynamics



6. Theory: Geometric Intuition



7. Practical Takeaway

- Data selection jointly with model capacity and compute budget.**
- Small Model / Short Budget:** prefer high-likelihood data
- Transition Regime:** checkpoint-sensitive, mixed behavior
- Large Model / Longer Budget:** low-likelihood data can unlock stronger reasoning

We identify a capacity-dependent Fast-Fit / Slow-Gain phenomenon and provide both empirical and theoretical explanations for when high-likelihood or low-likelihood supervision is most effective.